Miso Lee

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RESEARCH INTEREST

3D Vision: Visual Localization, Camera Pose Estimation, Multi-view Geometry, Human-Scene Reconstruction

EDUCATION

Sungkyunkwan University — Suwon, South Korea
Ph.D. Student in Artificial Intelligence *(Advisor: Jae-Pil Heo)* — *Mar. 2023 – Present*

Sungkyunkwan University — Suwon, South Korea
B.S. in Mechanical Engineering & Computer Science and Engineering *(Double Major)* — *Mar. 2019 – Feb. 2023*

PUBLICATIONS

Disambiguating 2D-3D Correspondences in Gaussian Splatting-based Feature Fields for Visual Localization
Miso Lee, Sangeek Hyun, Yerim Jeon, and Jae-Pil Heo
Advances in Neural Information Processing Systems (NeurIPS) *2026*

White Aggregation and Restoration for Few-shot 3D Point Cloud Semantic Segmentation
Jiyun Im, SuBeen Lee, Miso Lee, and Jae-Pil Heo
European Conference on Computer Vision (ECCV) *2026*

Long-term Pre-training for Temporal Action Detection with Transformers
Jihwan Kim, Miso Lee, and Jae-Pil Heo
Pattern Recognition *2026*

Masking Matters: Unlocking the Spatial Reasoning Capabilities of LLMs for 3D Scene-Language Understanding
Yerim Jeon, Miso Lee, WonJun Moon, and Jae-Pil Heo
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) *2026*

Temporally Consistent Long-Term Memory for 3D Single Object Tracking
Jaejoon Yoo, SuBeen Lee, Yerim Jeon, Miso Lee, and Jae-Pil Heo
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Findings *2026*

Prediction-Feedback DETR for Temporal Action Detection
Jihwan Kim, Miso Lee, Cheol-Ho Cho, Jihyun Lee, and Jae-Pil Heo
AAAI Conference on Artificial Intelligence (AAAI) *2025*

Activating Self-Attention for Multi-Scene Absolute Pose Regression
Miso Lee, Jihwan Kim, and Jae-Pil Heo
Advances in Neural Information Processing Systems (NeurIPS) *2024*

Self-Feedback DETR for Temporal Action Detection
Jihwan Kim, Miso Lee, and Jae-Pil Heo
IEEE/CVF International Conference on Computer Vision (ICCV) *2023*

EXPERIENCES

Research Intern — *May 2026 – Aug. 2026*
NAVER Cloud, Video Generation Team (4D Group)
- Developed feature fusion and metric scale calibration modules for human-scene reconstruction.
- Paper in preparation: *Human-to-Human Registration: Plug-in Scale Calibration for Human-Scene Reconstruction*.

SELECTED PROJECTS

3D Perception for Autonomous Pallet Stacker — *Jun. 2024 – Mar. 2025*
- Co-led a project on LiDAR- and stereo RGB-based 3D perception for automated pallet handling.
- Developed 3D unprojection error calibration and registration-based pose estimation modules.

Efficient 3D Reconstruction via Distributed Processing and Merging — *May 2022 – Nov. 2023*
- Participated as a project member in an ETRI-sponsored project.
- Investigated image clustering strategies for large-scale structure-from-motion.

HONORS AND AWARDS

Merit Fellowship
Nationally competitive, Ilun Science and Technology Foundation *Dec. 2023 – Aug. 2026*

Merit Scholarship
Half-tuition, Department of Artificial Intelligence, Sungkyunkwan University *Mar. 2023 – Sep. 2025*